



Bioinformatics for dummies

Computational environment

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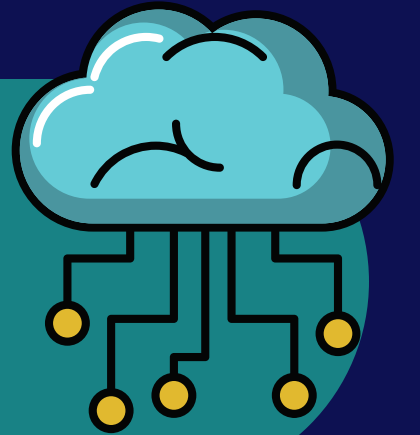


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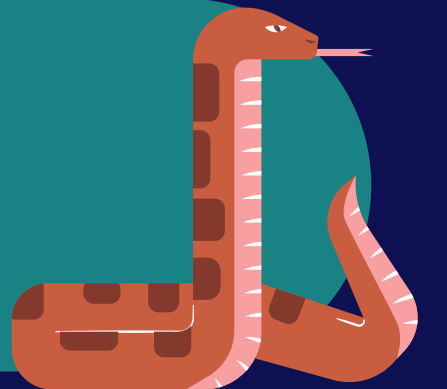
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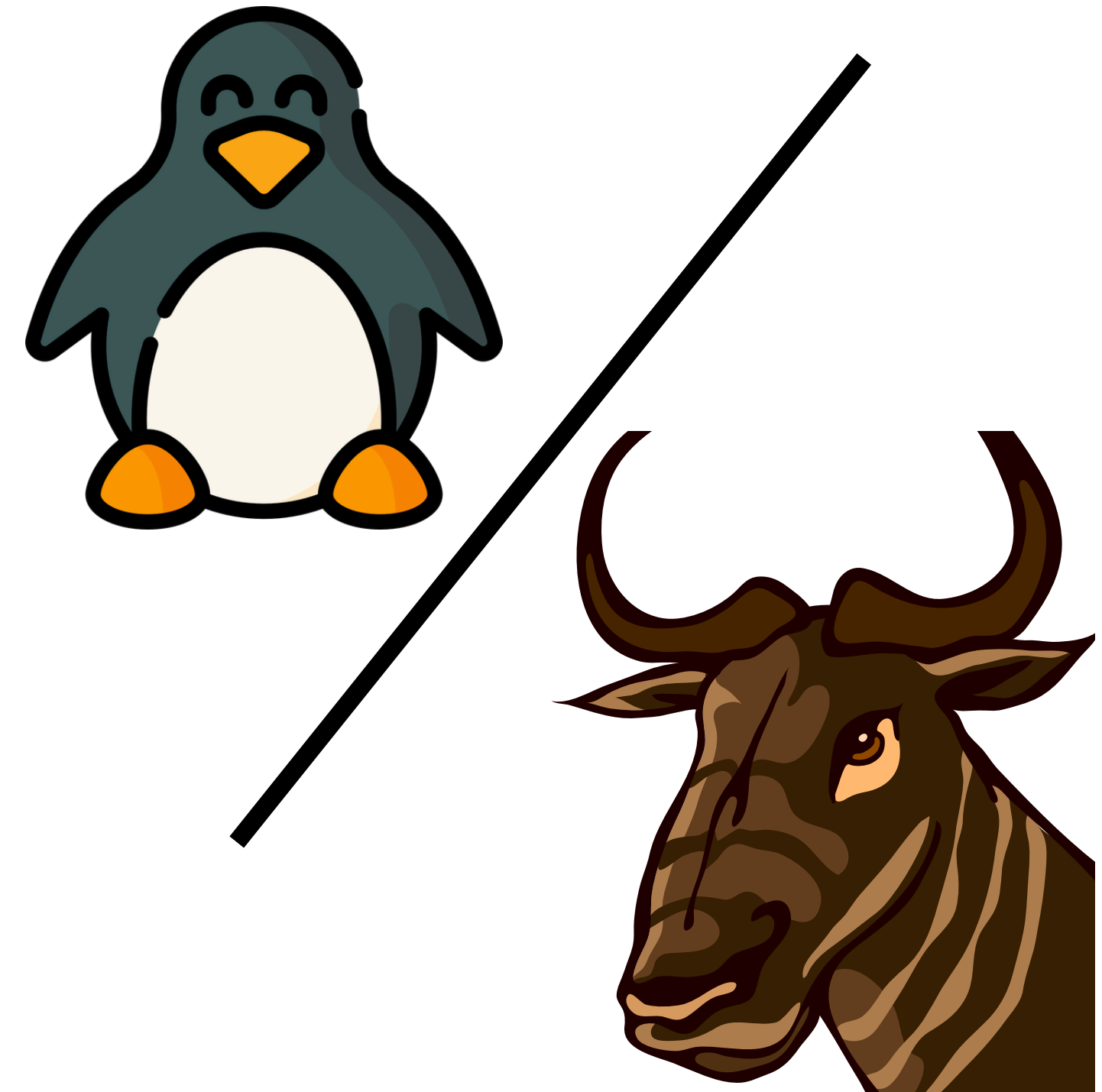


SOMETHING ABOUT LINUX

Linux is not an OS

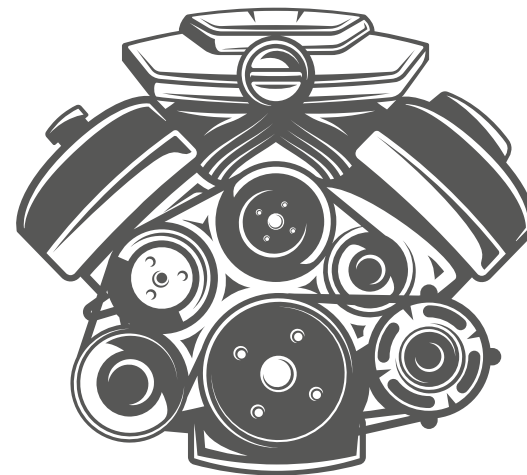
Linux is essentially the kernel of a set of operating systems commonly referred to as GNU/LINUX.

The kernel is the central piece of software that allows high-level software to communicate with the available hardware.



Unity is strength

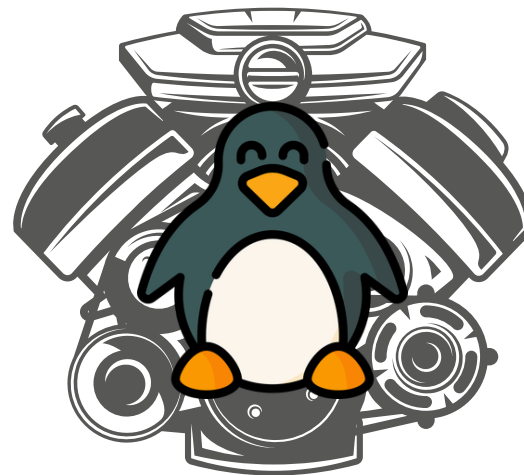
The open science



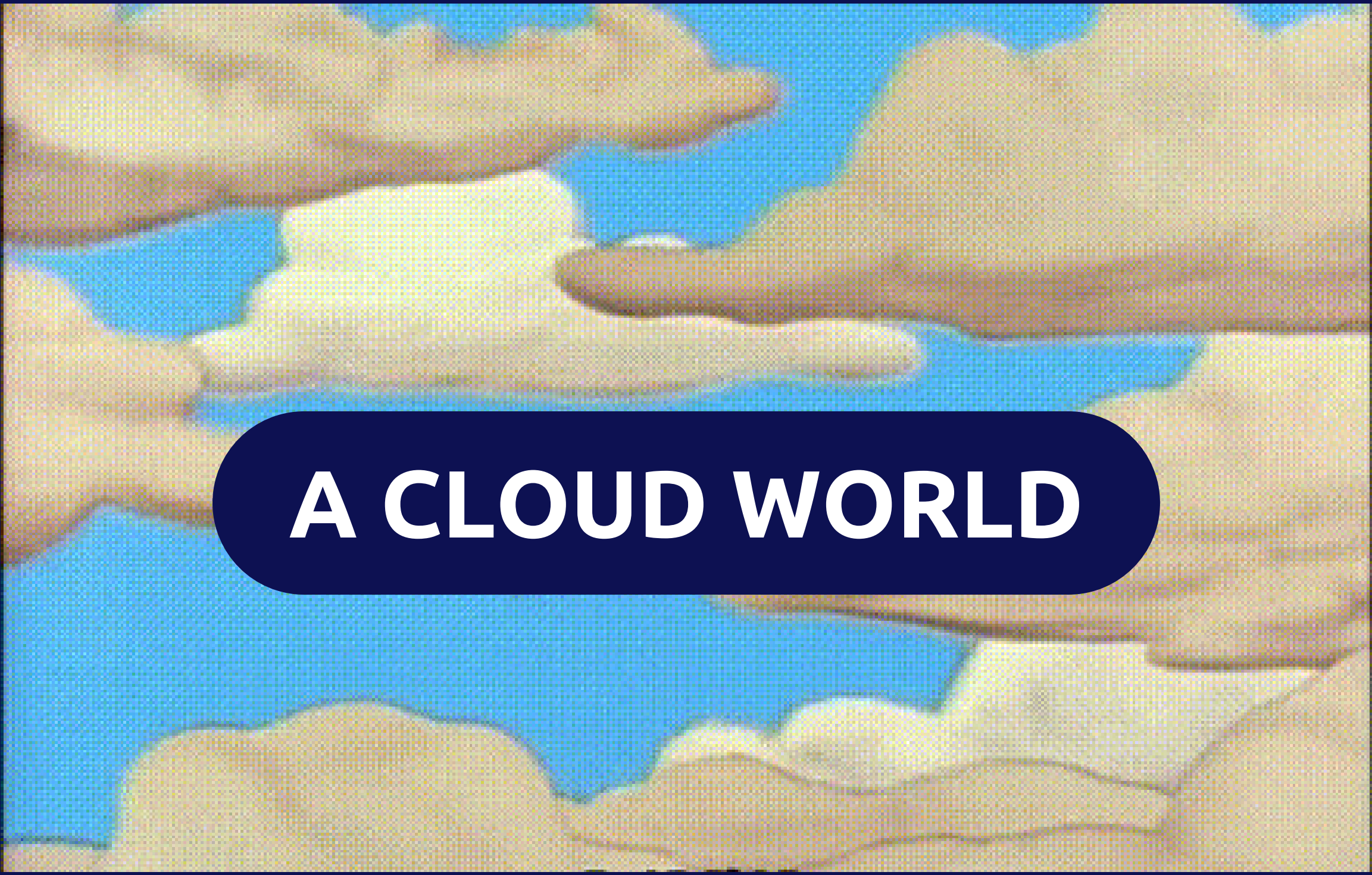
- As in a car, the operating systems have many parts.
- Linux based systems have parts from multiple authors instead of a single manufacturer producing the entire car.

Unity is strength

The open science



- As in a car, the operating systems have many parts.
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A CLOUD WORLD

how popular is linux?

Linux is INDISPENSABLE today.

Linux is a fully customizable layman

Microsoft (example) is VERY stable but NOT customizable

GNU/LINUX systems are the least used on desktops (< 2%)

You have to install it

Low compatibility with proprietary developments

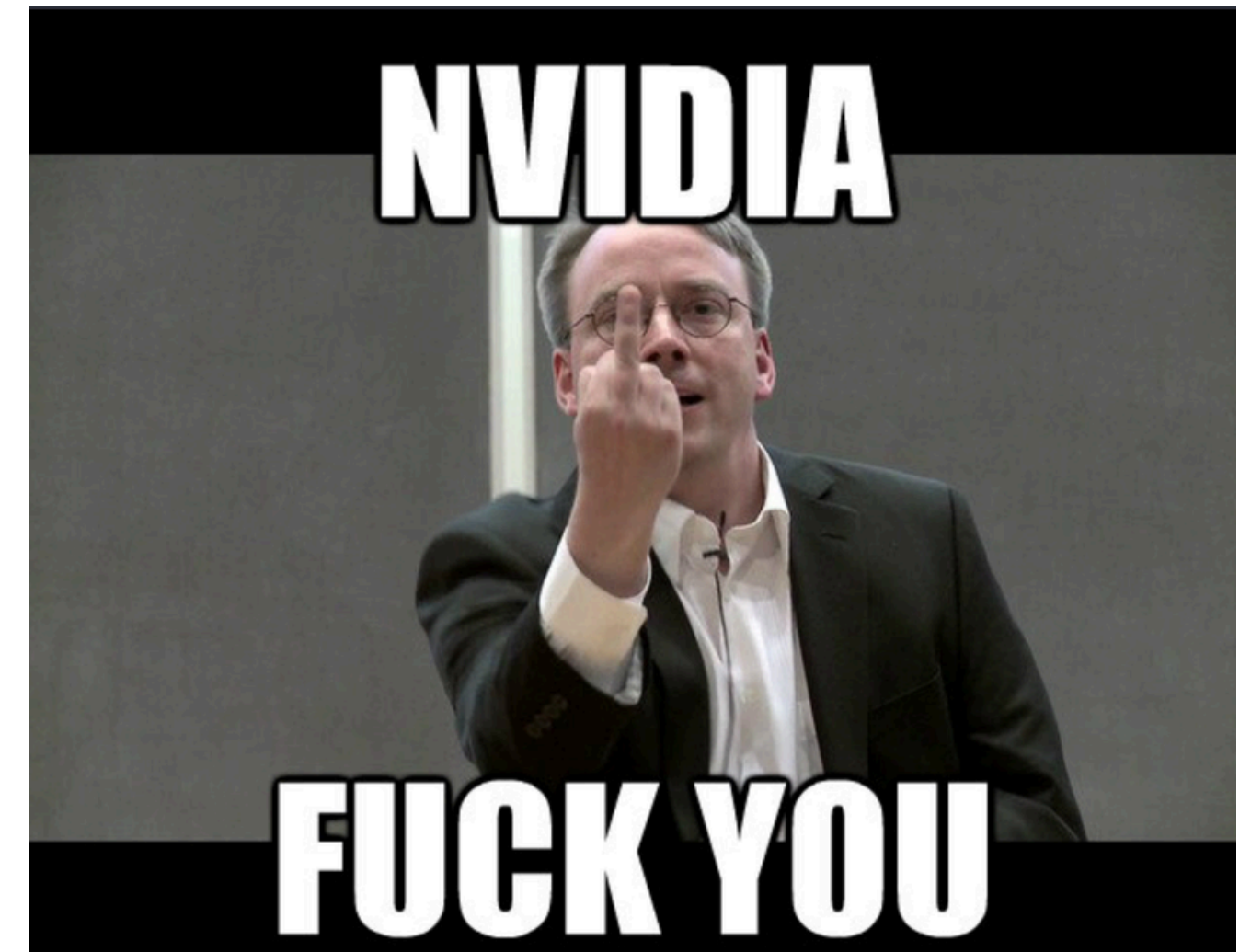
Low compatibility with hardware

Unstable but free drivers

No proprietary operating system has been able to meet the needs of servers.

Linux, being free and customizable, has taken control in enterprise computing (> 60%)

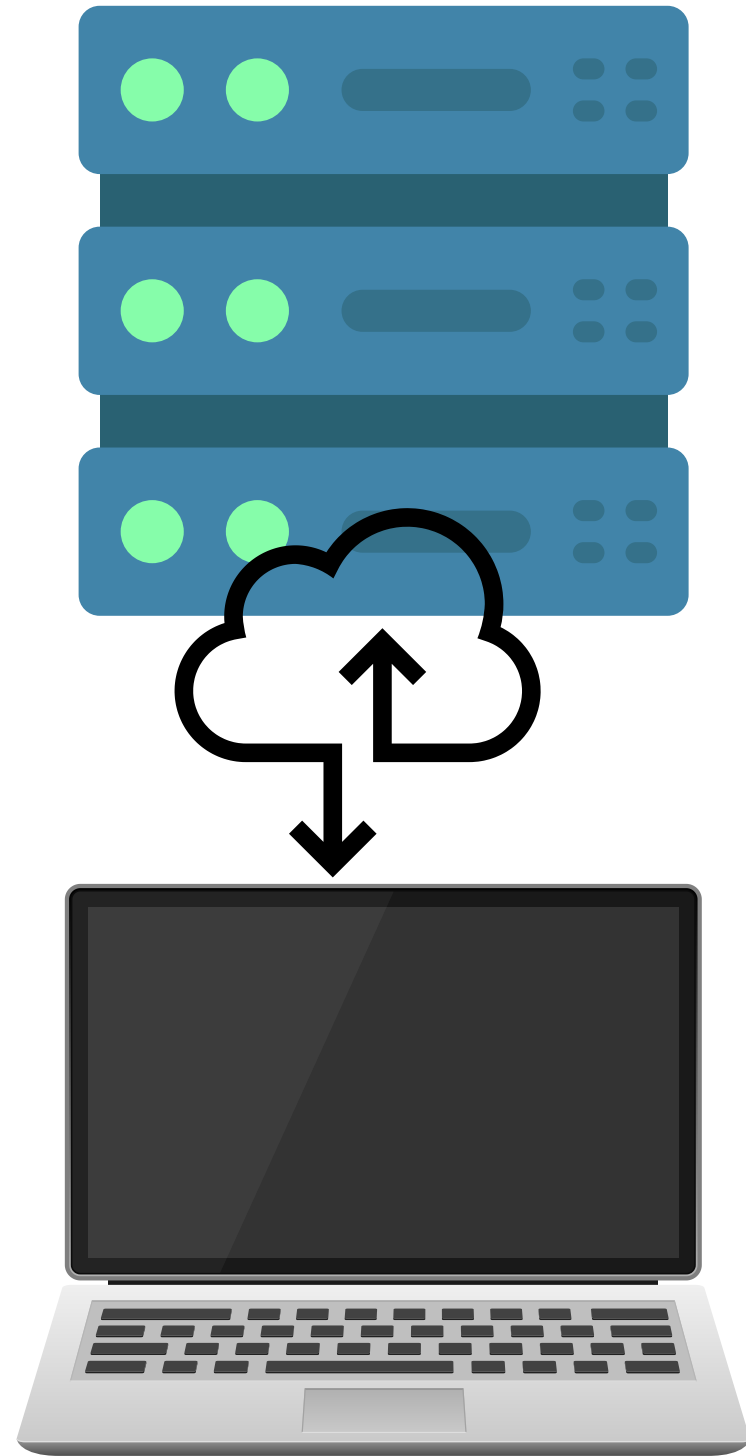
From servers leap to smartphones, PDAs, embedded systems, routers, IoT



<https://www.youtube.com/shorts/tQIdxbWhHSM>

Linux in bioscience

we need servers



Normally, a personal computer does not have the capabilities to **process large volumes of data**.

It is necessary to use cloud services:

- GCP (google)
- AWS (amazon)
- AZURE (microsoft)
- AutoHOST

Most cloud services run on GNU/LINUX systems and offer tools that assume the availability of a terminal (**bash**) in a local environment.

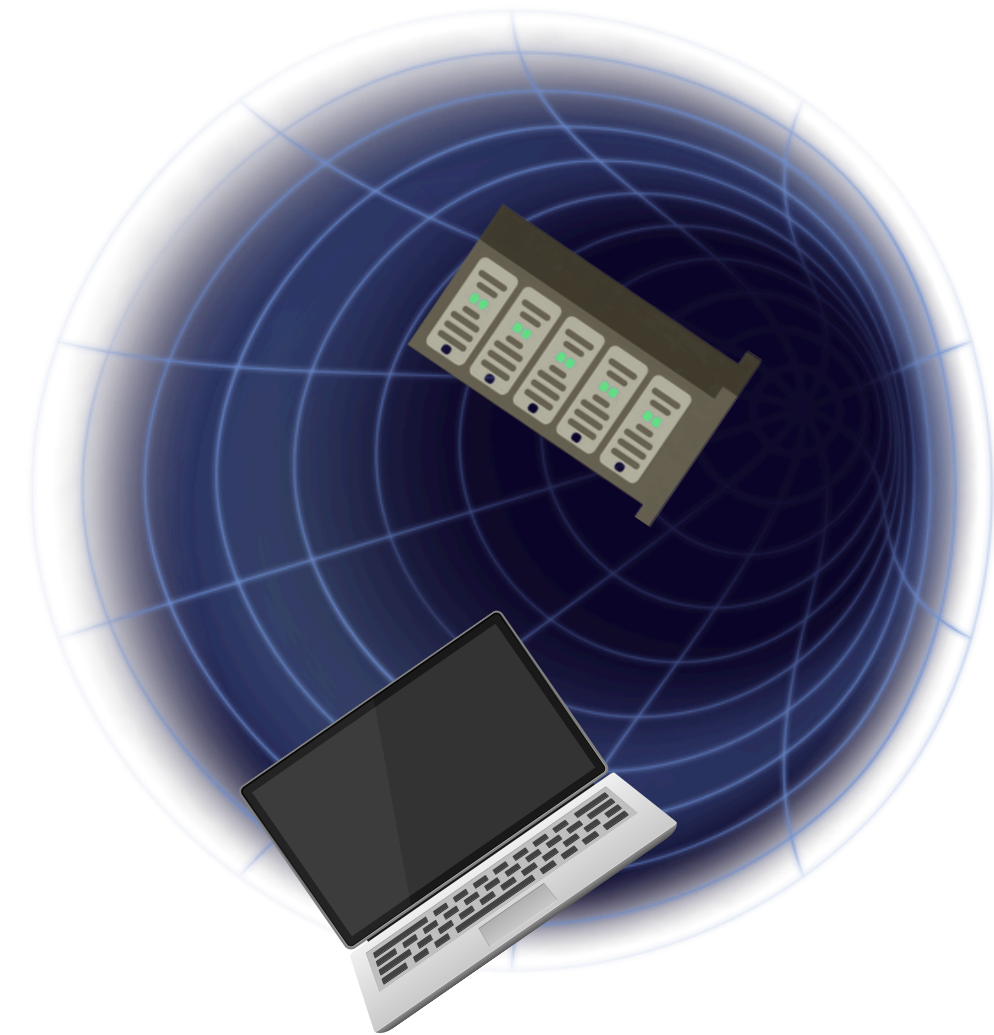
Managing multiple tasks and process **queues** is easier in LINUX environments.

Maintaining extensive and long-lasting processes is more **efficient** on observed and dedicated hardware such as **servers**.

BREAKING THE DISTANCE: THE SECURE SHELL

The SSH wormhole

Connecting to the server



SSH is a protocol that allows secure remote access over the network

- Provides an encrypted channel for communication between your local computer and the server to be private.
- It is mainly operated through the terminal (bash)

As the processing of large volumes of data takes place on dedicated servers and not on local machines, SSH is the indispensable "bridge" to execute commands and manage files remotely.

THE PENGUIN HISTORY

Unix to Linux



debian



GNOME™

1983
GNU project

1986
Free Software
Foundation

1990
CYGNUM

1991
Linux kernel

1993
Debian project

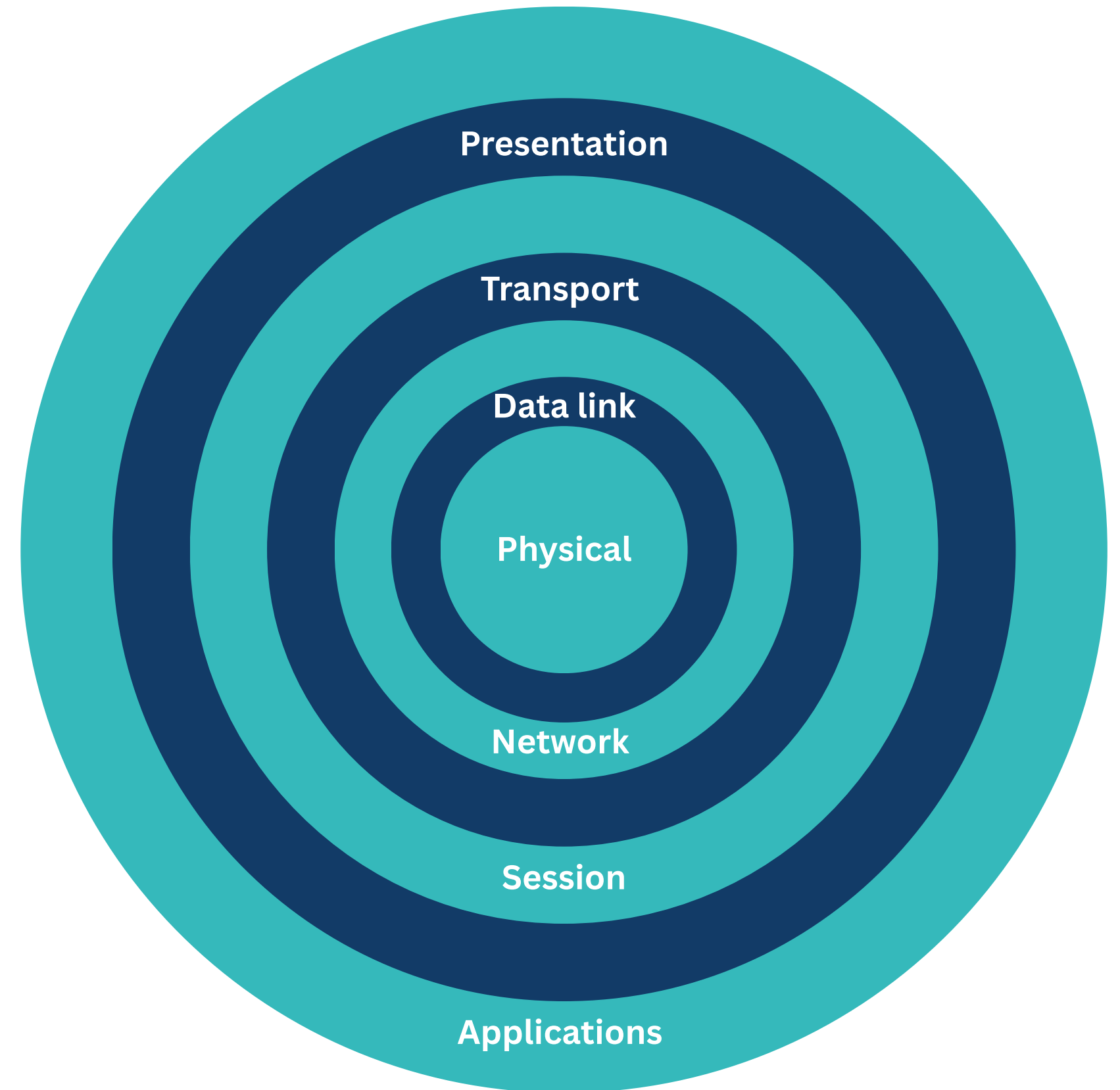
1996
Gnome desktop

**COMPUTERS ARE LIKE
ONIONS**

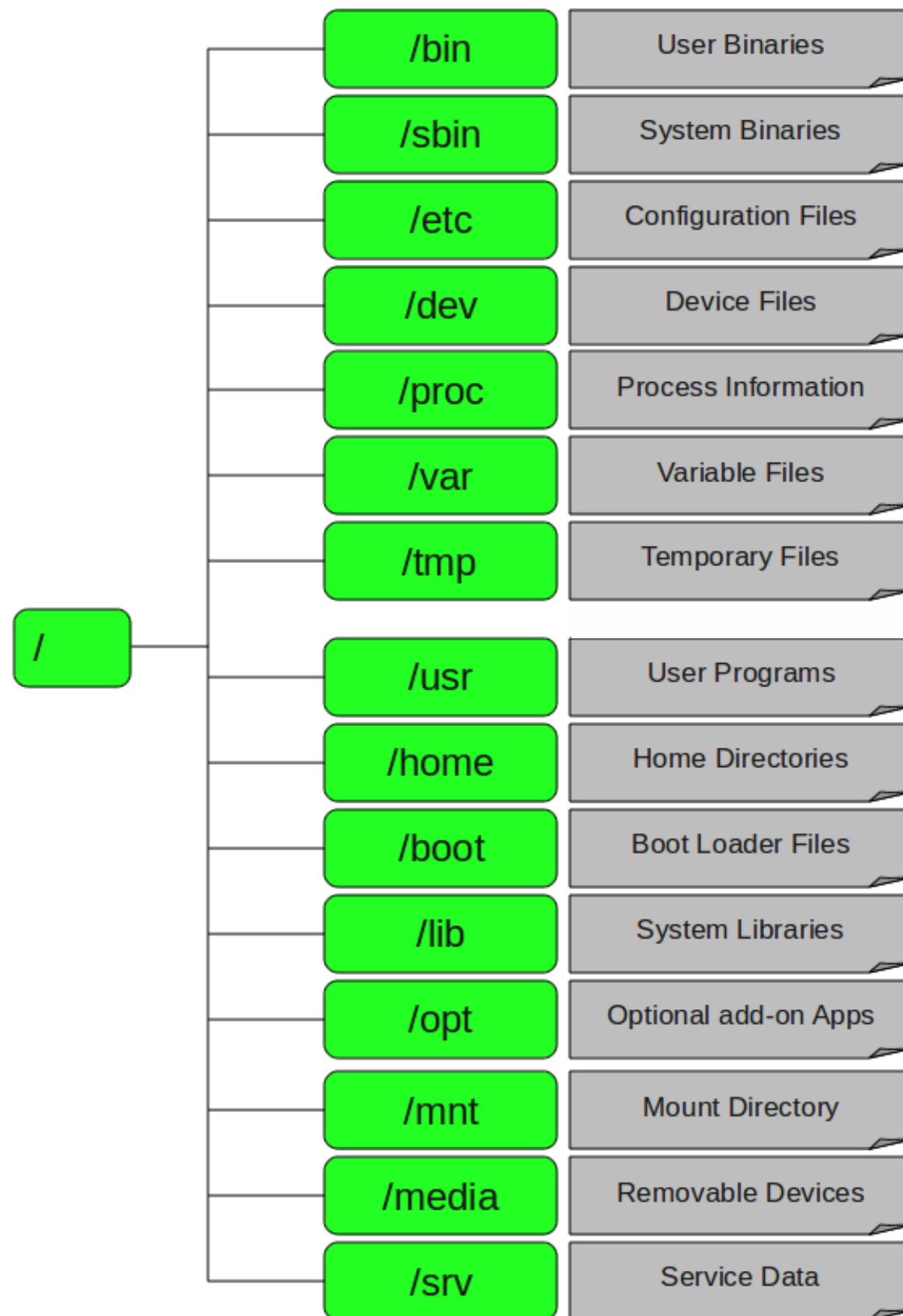
The layers of an OS

The system in general has different parts:

- **Kernel (This is LINUX!)**
- **Core applications (and this is GNU)**
- High level applications
- Shell CLI/graphics



The linux dirs



- In linux everything is directories, this structure allows a simple manipulation of the OS structure.
- Access to directories is handled by the user's permission manager, if a user has admin permissions he CAN MODIFY ANY PART OF THE SYSTEM.

LINUX AND ITS FLAVOURS

Distributions



R VS PYTHON

Which one to choose?



- Specialized packages are available
- Developed for statistical analysis
- Difficult to learn
- Slow
- Poor integration with complex systems



- Lacks specialized bioinformatics packages
- Optimization on large data sets
- VERY VERY easy to learn
- Integration with complex hardware

Thanks !!!!!